Contents

Preface 9

1 Functions 19

1.1 Functions and Their Graphs 19
1.2 Combining Functions; Shifting and Scaling Graphs 32
1.3 Trigonometric Functions 39
1.4 Graphing with Software 47
Questions to Guide Your Review 51
Practice Exercises 52
Additional and Advanced Exercises 53
Technology Application Projects 55

2 Limits and Continuity 56

2.1 Rates of Change and Tangent Lines to Curves 56
2.2 Limit of a Function and Limit Laws 63
2.3 The Precise Definition of a Limit 74
2.4 One-Sided Limits 83
2.5 Continuity 90
2.6 Limits Involving Infinity; Asymptotes of Graphs 101
Questions to Guide Your Review 114
Practice Exercises 115
Additional and Advanced Exercises 116
Technology Application Projects 119

3 Derivatives 120

3.1 Tangent Lines and the Derivative at a Point 120
3.2 The Derivative as a Function 124
3.3 Differentiation Rules 133
3.4 The Derivative as a Rate of Change 142
3.5 Derivatives of Trigonometric Functions 152
3.6 The Chain Rule 158
3.7 Implicit Differentiation 166
3.8 Related Rates 171
3.9 Linearization and Differentials 180
Questions to Guide Your Review 192
Practice Exercises 192
Additional and Advanced Exercises 197
Technology Application Projects 200

4 Applications of Derivatives 201

4.1 Extreme Values of Functions on Closed Intervals 201
4.2 The Mean Value Theorem 209
4.3 Monotonic Functions and the First Derivative Test 215
4.4 Concavity and Curve Sketching 220
4.5 Applied Optimization 232
4.6 Newton's Method 244
4.7 Antiderivatives 249
Questions to Guide Your Review 259
Practice Exercises 259
Additional and Advanced Exercises 262
Technology Application Projects 265

5 Integrals 266

5.1 Area and Estimating with Finite Sums 266
5.2 Sigma Notation and Limits of Finite Sums 276
5.3 The Definite Integral 283
5.4 The Fundamental Theorem of Calculus 296
5.5 Indefinite Integrals and the Substitution Method 307
5.6 Definite Integral Substitutions and the Area Between Curves 314
Questions to Guide Your Review 324
Practice Exercises 325
Additional and Advanced Exercises 328
Technology Application Projects 331

6 Applications of Definite Integrals 332

6.1 Volumes Using Cross-Sections 332
6.2 Volumes Using Cylindrical Shells 343
6.3 Arc Length 351
6.4 Areas of Surfaces of Revolution 356
6.5 Work and Fluid Forces 362
6.6 Moments and Centers of Mass 371
Questions to Guide Your Review 383
Practice Exercises 384
Additional and Advanced Exercises 386
Technology Application Projects 387

7 Transcendental Functions 388

7.1 Inverse Functions and Their Derivatives 388
7.2 Natural Logarithms 396
7.3 Exponential Functions 404
7.4 Exponential Change and Separable Differential Equations 415
7.5 Indeterminate Forms and L'Hôpital's Rule 425
7.6 Inverse Trigonometric Functions 434
7.7 Hyperbolic Functions 446

7.8 Relative Rates of Growth 454
Questions to Guide Your Review 459
Practice Exercises 460
Additional and Advanced Exercises 463

8 Techniques of Integration 465

8.1 Using Basic Integration Formulas 465
8.2 Integration by Parts 470
8.3 Trigonometric Integrals 478
8.4 Trigonometric Substitutions 484
8.5 Integration of Rational Functions by Partial Fractions 489
8.6 Integral Tables and Computer Algebra Systems 497
8.7 Numerical Integration 503
8.8 Improper Integrals 512
Questions to Guide Your Review 523
Practice Exercises 524
Additional and Advanced Exercises 526
Technology Application Projects 529

9 Infinite Sequences and Series 530

9.1 Sequences 530
9.2 Infinite Series 543
9.3 The Integral Test 553
9.4 Comparison Tests 559
9.5 Absolute Convergence; The Ratio and Root Tests 564
9.6 Alternating Series and Conditional Convergence 571
9.7 Power Series 578
9.8 Taylor and Maclaurin Series 589
9.9 Convergence of Taylor Series 594
9.10 Applications of Taylor Series 601
Questions to Guide Your Review 610
Practice Exercises 611
Additional and Advanced Exercises 613
Technology Application Projects 615

10 Parametric Equations and Polar Coordinates 616

10.1 Parametrizations of Plane Curves 616
10.2 Calculus with Parametric Curves 625
10.3 Polar Coordinates 634
10.4 Graphing Polar Coordinate Equations 638
10.5 Areas and Lengths in Polar Coordinates 642
10.6 Conic Sections 647
10.7 Conics in Polar Coordinates 655
Questions to Guide Your Review 661
Practice Exercises 662
Additional and Advanced Exercises 664
Technology Application Projects 666

11 Vectors and the Geometry of Space 667

11.1 Three-Dimensional Coordinate Systems 667
11.2 Vectors 672
11.3 The Dot Product 681
11.4 The Cross Product 689
11.5 Lines and Planes in Space 695
11.6 Cylinders and Quadric Surfaces 704
Questions to Guide Your Review 710
Practice Exercises 710
Additional and Advanced Exercises 712
Technology Application Projects 715

12 Vector-Valued Functions and Motion in Space 716

12.1 Curves in Space and Their Tangents 716
12.2 Integrals of Vector Functions; Projectile Motion 725
12.3 Arc Length in Space 734
12.4 Curvature and Normal Vectors of a Curve 738
12.5 Tangential and Normal Components of Acceleration 744
12.6 Velocity and Acceleration in Polar Coordinates 750
Questions to Guide Your Review 754
Practice Exercises 755
Additional and Advanced Exercises 757
Technology Application Projects 758

13 Partial Derivatives 759

13.1 Functions of Several Variables 759
13.2 Limits and Continuity in Higher Dimensions 767
13.3 Partial Derivatives 776
13.4 The Chain Rule 788
13.5 Directional Derivatives and Gradient Vectors 798
13.6 Tangent Planes and Differentials 806
13.7 Extreme Values and Saddle Points 816
13.8 Lagrange Multipliers 825
13.9 Taylor's Formula for Two Variables 835
13.10 Partial Derivatives with Constrained Variables 839
Questions to Guide Your Review 843
Practice Exercises 844
Additional and Advanced Exercises 847
Technology Application Projects 849

14 Multiple Integrals 850

14.1 Double and Iterated Integrals over Rectangles 850
14.2 Double Integrals over General Regions 855
14.3 Area by Double Integration 864
14.4 Double Integrals in Polar Form 867
14.5 Triple Integrals in Rectangular Coordinates 874
14.6 Applications 884
14.7 Triple Integrals in Cylindrical and Spherical Coordinates 894
14.8 Substitutions in Multiple Integrals 906
Questions to Guide Your Review 916
Practice Exercises 916
Additional and Advanced Exercises 919
Technology Application Projects 921

15 Integrals and Vector Fields 922

15.1 Line Integrals of Scalar Functions 922
15.2 Vector Fields and Line Integrals: Work, Circulation, and Flux 929
15.3 Path Independence, Conservative Fields, and Potential Functions 942
15.4 Green's Theorem in the Plane 953
15.5 Surfaces and Area 965
15.6 Surface Integrals 975
15.7 Stokes' Theorem 985
15.8 The Divergence Theorem and a Unified Theory 998
Questions to Guide Your Review 1011
Practice Exercises 1011
Additional and Advanced Exercises 1014
Technology Application Projects 1015

16 First-Order Differential Equations 1016

16.1 Solutions, Slope Fields, and Euler's Method 1016
16.2 First-Order Linear Equations 1024
16.3 Applications 1030
16.4 Graphical Solutions of Autonomous Equations 1036
16.5 Systems of Equations and Phase Planes 1043
Questions to Guide Your Review 1049
Practice Exercises 1049
Additional and Advanced Exercises 1051
Technology Application Projects 1052

17 Second-Order Differential Equations (Online)

17.1 Second-Order Linear Equations
17.2 Nonhomogeneous Linear Equations
17.3 Applications
17.4 Euler Equations
17.5 Power-Series Solutions

Appendices AP-1

A.1 Real Numbers and the Real Line AP-1
A.2 Mathematical Induction AP-6
A.3 Lines, Circles, and Parabolas AP-9
A.4 Proofs of Limit Theorems AP-19
A.5 Commonly Occurring Limits AP-22
A.6 Theory of the Real Numbers AP-23
A.7 Complex Numbers AP-26
A.8 Probability AP-34
A.9 The Distributive Law for Vector Cross Products AP-47
A.10 The Mixed Derivative Theorem and the Increment Theorem AP-49

Answers to Odd-Numbered Exercises A-1

Credits C-1

Applications Index AI-1

Subject Index SI-1

A Brief Table of Integrals T-1

Basic Algebra Formulas B-1

Limits, Differentiation, and Integration Rules L-2